

Dongwook Kwon

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Experience

Suresoft Technologies

Seongnam-si, Korea

AI ENGINEER, INTELLIGENT AGENT TECHNOLOGY TEAM

Dec. 2025 - Present

- Developing an AI validation system utilizing AI agent-based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University

Seoul, Korea

RESEARCH ASSISTANT

Aug. 2021 - Dec. 2025

- Researched deep-learning algorithms for anomaly detection in complex time-series data, focusing on critical cyber-physical systems including nuclear power generation environments.
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.

LG Electronics Canada (with University of Toronto)

Toronto, ON, Canada

VISITING RESEARCHER

Jan. 2025 - Jul. 2025

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

Qualcomm Institute, University of California San Diego

San Diego, CA, USA

AI DEVELOPMENT PROJECT PARTICIPANT, SUMMER PROGRAM

Jul. 2022 - Aug. 2022

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

Education

Kwangwoon University

Seoul, Korea

MASTER OF SCIENCE IN COMPUTER ENGINEERING

Mar. 2024 - Feb. 2026

- GPA: 4.33/4.5 (98.1%)

University of Toronto

Toronto, ON, Canada

GRADUATE RESEARCH PROGRAM IN MECHANICAL AND INDUSTRIAL ENGINEERING

Jan. 2025 - Jul. 2025

- GPA: 3.68/4.0

Kwangwoon University

Seoul, Korea

BACHELOR OF SCIENCE IN ELECTRONICS AND COMMUNICATIONS ENGINEERING

Mar. 2019 - Feb. 2024

- GPA: 3.41/4.5 (88.1%)

Honors & Awards

MAJOR AWARDS

Aug. 2022 **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego

United States

Dec. 2021 **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries

South Korea

ACADEMIC RECOGNITION

Nov. 2025 **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers

South Korea

Nov. 2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024)

South Korea

Nov. 2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering

South Korea

COMPETITIONS

Dec. 2025 **Silver Award**, 23rd Embedded Software Competition (KIISE)

South Korea

Dec. 2022 **Bronze Award**, Hanium ICT Mentoring Competition — FKII

South Korea

Sep. 2022 **Excellence Award**, 18th Kwangwoon ICT Exhibition (KWIX) — Kwangwoon University

South Korea

Oct. 2021 **Grand Prize**, MY (Multi-Y) Capstone Design Competition — Kwangwoon University

South Korea

Funded Research Projects

LG Electronics Canada AGENTIC AI EVALUATION FRAMEWORK Developed an evaluation framework for agentic AI systems using LLMs to assess performance.	Toronto, ON, Canada Jan. 2025 - Present
Sejong University AI-BASED INTRUSION DETECTION AND ATTACK PACKET CRAFTING TECHNOLOGIES FOR APR1400 Developed an AI defense system to protect nuclear power plants from cyber attacks.	Seoul, Korea Jul. 2022 - Present
Ministry of Science and ICT QUADRUPED ROBOT USING VISION SLAM WITH A DEPTH CAMERA Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.	Seoul, Korea Apr. 2022 - Nov. 2022
Qualcomm Institute, University of California San Diego PERSONALITY-BASED DRUG ADDICTION PREDICTION SYSTEM Developed a personality-based drug addiction prediction system using a machine learning model.	San Diego, CA, USA Jul. 2022 - Aug. 2022
Kookmin University DEVELOPMENT OF ACTIVE KILL-SWITCH AND BIOMARKER-BASED DEFENSE SYSTEM FOR LIFE-THREATENING IOT MEDICAL DEVICES Developed a security system for heart implant devices using an anomaly detection algorithm.	Seoul, Korea Sep. 2021 - Dec. 2022
Ministry of Oceans and Fisheries DEVELOPMENT OF A SMART LOGISTICS SYSTEM USING COMPUTER VISION Developed a logistics detection system incorporating location tracking and stack-layer identification for containers.	Seoul, Korea Apr. 2021 - Nov. 2021
Kwangwoon University SMART KITCHEN IOT SYSTEM UTILIZING COMPUTER VISION AND DEEP LEARNING Developed a cloud IoT platform featuring a system for classifying groceries in the fridge and an OCR receipt program.	Seoul, Korea May 2021 - Oct. 2021

Publications

JOURNAL ARTICLES

- Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, “Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise,” *Electronics*, vol. 14, no. 5, p. 890, 2025.
- D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, “Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments,” under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

CONFERENCE PAPERS

- R. Chang[†], **D. Kwon**[†], J. Lee[†], and N. Verma, “CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades,” in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) — Industry Track (Poster)*, 2026.
[†]Equal contribution.
- J. Lee[†], R. Chang[†], **D. Kwon**[†], H. Singh, and N. Verma, “GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems,” in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025.
[†]Equal contribution.
- D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, “Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers,” in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. **Best Poster Award — Silver.**
- D. Kwon**, Y. Kang, and C. Park, “Enhancing medical device security with GNN-GRU anomaly detection model,” in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. **Outstanding Paper Award.**

Patents

Apr 2026	Method for Implantable Medical Device to Detect Anomaly in Real Time , KR 10-2961984	Republic of Korea
Oct 2025	System for Anomaly Detection Using a Collaborative Multi-AI Determination Module Based on LLM , KR 10-2025-0160217 (Pending)	Republic of Korea
Aug 2025	GPT API-Based Network Anomaly Detection , KR 10-2848814	Republic of Korea
Aug 2024	Categorical Data of Networks Based Network Anomaly Detection Apparatus and Method Thereof , KR 10-2024-0110559 (Pending)	Republic of Korea

Leadership Experience

Kwangwoon International Student Association

PRESIDENT

- Lead intercultural programs and student engagement initiatives.

Kwangwoon Broadcasting Center (KWBC)

DIRECTOR OF FILMING AND EDITING

- Oversaw media production and content delivery.

Dept. of Electronics and Communications Engineering

CLASS REPRESENTATIVE

- Represented student interests in academic affairs.

Amateur Radio Club

EXECUTIVE MEMBER

- Organized radio workshops and technical exhibitions.